PROBLEM STATEMENT 04 • TEAM NACHOS • G M JAGATH SAJJAN

VoltC: Real-Time Building Energy Intelligence Twin

Computer vision monitoring to detect and reduce commercial energy waste.

ARCHITECTURE

System Workflow

1. Video Feed

CCTV sector clips and live USB webcams.

2. Vision Engine

YOLOv8 headcount and room brightness thresholding.

3. Rules Core

State classification, server load filters, and cost tracking.

4. GenAI Action

Gemini and Groq fallback automated action plans.

LOGIC

Room State Classification

EFFICIENT

Lights off or low energy draw.

OCCUPIED

People present with active load.

INEFFICIENT

High power draw for single occupant.

WASTE

Lights on with zero occupants.

VALIDATION

Classification Performance

Accuracy	**85.3%**
F1 Score	**0.818**
ROC-AUC	**0.915**

Evaluation Notes

Tested across varying natural sunlight glare and night surveillance conditions.

Decision threshold set at 0.43 to minimize false positive waste alarms.

CALCULATIONS

Mathematical Formulas

Energy Wasted (kWh)

`(Watts × Waste Hours) / 1000`

Cost Wasted (INR)

`kWh Wasted × Tariff Rate (₹9.50)`

Carbon Emitted (kg)

`kWh Wasted × Emission Factor (0.71)`

METHODOLOGY

Algorithm & YOLO Selection

Why YOLOv8 over Others?

Real-time inference speed needed for live CCTV streams.

Lightweight bounding box detection without heavy cloud GPU bottlenecks.

How the Algorithm Works

Extracts grayscale frame brightness against per-room thresholds.

Filters 24/7 server racks as infrastructure; evaluates switchable load waste.

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Thank You • Questions

Live Demo on Streamlit Cloud